QUANTITATIVE FINANCE ASSUMPTION REGISTER

Assumptions, design decisions, modelling conventions, limitations, and open questions

Field	Details
As of	28 October 2026
Project root	C:\DESKTOP-1\GITHUB_REPOS\Finance_Project_Updated
Environment	Windows PowerShell; Python 3.11.5 in .venv
Purpose	Record the assumptions and unresolved decisions that may affect research results and future development.

How to read this register

Established means explicitly reflected in the documented implementation/results. **Working assumption** means a current simplifying choice, not necessarily the final choice. **Verify/decide** marks something to inspect in source code or decide before making stronger claims.

> Passing tests establishes expected software behavior for tested cases; it does not prove that every financial modelling assumption is economically realistic.

1. Project Scope and Development Principles

ID / topic	Assumption or decision	Status / implication
P-01 — Research platform	Reusable quantitative-finance research and backtesting platform, not a one-off notebook.	Established; keep data, strategies, simulation, metrics, validation, and charts modular.
P-02 — Validate before ML	Establish reliable baselines and validation before adding machine learning.	Established; avoid adding complexity before the evaluation framework is dependable.
P-03 — Incremental development	Implement incrementally and validate meaningful changes before moving on.	Established; run targeted tests, full tests, and Ruff after relevant changes.
P-04 — Naming	Final project name remains undecided; distribution name is research-platform and import package is project_structure.	Established; avoid hardcoding an assumed final name.
P-05 — Git	No Git repository has been initialized.	Established current state; reconsider deliberately later.
P-06 — Python version	Use Python 3.11 due to earlier package/build compatibility issues.	Established; continue using the project .venv.
P-07 — Operating environment	Windows PowerShell is the main development environment.	Established; portability has not been independently verified.
P-08 — Project objective	Build a defensible advanced DS/ML and quantitative-finance portfolio project.	Established; prioritize research rigor over a high headline return.

2. Data Source and Data Handling

ID / topic	Assumption or decision	Status / implication
D-01 — Provider	Yahoo Finance via yfinance.	Provider availability, revisions, symbols, and adjusted-price conventions affect reproducibility.

ID / topic	Assumption or decision	Status / implication
D-02 — Universe	SPY, QQQ, IWM, EFA, EEM, TLT, IEF, GLD, VNQ, DBC.	Results are conditional on this particular ETF universe.
D-03 — ETF proxies	ETFs proxy broad equity, bond, gold, real-estate, and commodity exposures.	ETFs have fees, tracking error, benchmark definitions, and changing holdings.
D-04 — Snapshot period	Documented refresh used start='2010-01-01'; prices: 2010-01-04 to 2026-10-27.	A future refresh can change history and the final date.
D-05 — Return field	Daily simple returns are inferred from price/return row counts; exact source field needs confirmation.	Verify adjusted close versus close and distribution/corporate-action treatment.
D-06 — Date alignment	Returns aligned on common dates across all configured assets.	Simplifies portfolio calculations but can discard dates not shared by all assets.
D-07 — Row counts	4,230 price rows per asset and 4,229 aligned return observations.	One fewer return observation than prices is expected.
D-08 — Pipeline default	Default start is 2020-01-01; pass start='2010-01-01' to request the longer history.	Refreshes may overwrite raw and processed CSVs; back up important snapshots.
D-09 — Latest date	Last date is the latest returned by the provider, not necessarily a finalized bar for every asset.	Record extraction time and provider settings.
D-10 — Validation	Shared loader validates files/date indices and aligns returns.	Document missing-data behavior; avoid silent filling unless intentional.
D-11 — Calendars	ETFs may have different trading calendars and holidays.	Timezone normalization and missing-date rules should be explicit.
D-12 — Survivorship/availability	The current ETF set is applied historically.	Assess inception dates, investability at the sample start, index changes, and survivorship/availability bias.
D-13 — Provider accuracy	Provider history is treated as sufficiently accurate for research.	Check adjustments and anomalous observations; preserve versioned snapshots for key experiments.

3. Return and Portfolio Accounting

ID / topic	Assumption or decision	Status / implication
R-01 — Frequency	Daily asset returns.	Annualization and risk metrics assume daily frequency.
R-02 — Constraints	Long-only weights sum to one for documented strategies.	No shorting or leverage; confirm optimizer constraints are consistent.
R-03 — Drift	Weights drift between rebalances as asset returns differ.	Portfolio is not reset to target weights every day.
R-04 — Turnover	L1 change in portfolio weights.	Verify whether reported turnover is one-way or two-way before interpreting traded notional.
R-05 — Cost model	Costs are proportional to measured turnover and charged on rebalance dates.	Simplified; clarify how turnover maps to traded notional.
R-06 — Cost/weights	Documented engine deducts cost without separately recalculating relative asset weights for the cost deduction.	Can differ slightly from full cash-and-holdings execution accounting.
R-07 — Stateful net return	net = (1 - cost) * (1 + gross) - 1.	Verify other backtest paths use identical accounting before comparing.
R-08 — Cost diagnostic	Sum of daily transaction-cost fractions is reported.	Not exactly cumulative portfolio-value loss attributable to costs.
R-09 — Capital	Normalized cumulative returns rather than a specified account balance.	No explicit dollar-level cash, integer shares, or rounding unless added.
R-10 — Financing	No explicit financing, margin, borrowing, or leverage model.	Not a complete live-trading implementation.
R-11 — Execution	No full model for spreads, market impact, liquidity, taxes, or execution delay.	May overstate implementable performance, especially for high-turnover strategies.
R-12 — Timing	Weights/return timing follows the current engine convention; exact alignment needs source audit.	Verify weights estimated using information through t-1 are applied to return on t, and when costs are charged.

ID / topic	Assumption or decision	Status / implication
R-13 — Input validity	Reject non-finite data, invalid weights, malformed dates, and returns below -100% in stateful simulation.	Keep validation consistent across engines and scripts.

4. Strategy Design and Optimization

ID / topic	Assumption or decision	Status / implication
S-01 — Benchmarks	Equal Weight, Minimum Variance, Maximum Sharpe.	Equal Weight is the simple baseline; optimized methods test whether estimation adds value.
S-02 — Lookback	252 trading days for the standard estimation window.	About one trading year; not universally optimal.
S-03 — Rebalance	21 trading days for the standard schedule; also test 5 and 63.	Candidate design choices, not universal optima.
S-04 — Minimum Variance	Uses historical covariance to minimize estimated portfolio variance.	Verify constraints, estimator, solver, regularization, and failure fallback.
S-05 — Maximum Sharpe	Uses historical return and covariance estimates to maximize estimated Sharpe.	Expected returns are noisy; document bounds, risk-free rate, solver, and fallback.
S-06 — Equal Weight role	Practical low-complexity benchmark, not claimed optimal.	May beat optimized portfolios when estimates are noisy.
S-07 — Full-sample weights	Used only for descriptive/in-sample comparison.	Not out-of-sample evidence because evaluation data also estimate weights.
S-08 — Concentration	Full-sample optimized allocations can concentrate heavily in IEF.	Add concentration, effective-number-of-assets, stability, and turnover diagnostics.
S-09 — Consistent constraints	Strategy constraints are intended to be consistent.	Verify bounds, sum-to-one, tolerances, and fallback rules in source.
S-10 — No guaranteed winner	Maximum Sharpe is not assumed to beat Equal Weight.	Strategy ranking is an empirical result, not a guaranteed property.

5. Performance and Risk Metrics

ID / topic	Assumption or decision	Status / implication
M-01 — Annualization	252 trading days per year.	Used for annualized return/volatility and Sharpe; document exact CAGR time convention.
M-02 — Risk-free rate	0% in documented baseline Sharpe calculations.	Simplification; not equivalent to an observed risk-free series.
M-03 — Rolling window	63 observations for rolling Sharpe and volatility.	About a quarter of trading days; early rows lack full window.
M-04 — Rolling volatility	Sample standard deviation (ddof=1) × sqrt(252).	Match this convention across reports and tests.
M-05 — Rolling Sharpe	Net returns and rolling sample standard deviation; annualized.	Costs affect rolling performance.
M-06 — Zero-volatility window	Undefined Sharpe rather than infinity.	Check cause before treating as data error.
M-07 — Drawdown	Cumulative wealth relative to prior peaks.	Confirm initial wealth and NaN handling in implementation.
M-08 — Multiple metrics	Use CAGR, total return, volatility, Sharpe, drawdown, and turnover together.	No single metric captures all risks and implementation costs.
M-09 — Diagnostics	Sum of turnover and cost fractions are reported.	Not equivalent to annual turnover, exact traded dollars, or exact wealth loss.
M-10 — Comparability	Compare only when dates, return conventions, and costs match.	Experiments have different protocols and should not be merged casually.

6. Out-of-Sample and Walk-Forward Design

ID / topic	Assumption or decision	Status / implication
W-01 — Initial lookback	252 days before standard out-of-sample evaluation.	Explains start of OOS period.
W-02 — Rebalance grid	5, 21, and 63 trading days.	Frequent, roughly monthly, and less frequent candidates.
W-03 — Cost grid	0, 10, and 25 bps.	Zero is a reference case; other costs are scenarios, not calibrated execution estimates.
W-04 — Same OOS period	All 27 robustness configurations share an OOS period.	Improves within-experiment comparability.
W-05 — Grid selection bias	Ranking 27 configurations by Sharpe can create selection bias.	Use separate holdout or nested walk-forward for final assessment.
W-06 — Selection history	Two calendar years of initial history.	Initial history length affects decisions.
W-07 — Candidate schedules	Expanding walk-forward compares 5, 21, and 63 days.	Selection can change with history.
W-08 — Historical-only selection	Stateful walk-forward uses Sharpe strictly before the test year.	Intended to prevent use of test-year outcomes in interval decisions.
W-09 — Tie break	Ties prefer the longer interval.	Deterministic rule favoring less frequent rebalancing in ties.
W-10 — Interval changes	Selected-interval changes at year boundaries force immediate rebalance.	Affects turnover/costs; not the only possible policy.
W-11 — Continuous holdings	Stateful simulation carries holdings and drift through time.	Avoids resetting portfolio to cash at each year.
W-12 — Audit fields	Target weights, applied weights, post-return drifted weights, interval-change flag.	Allows portfolio transitions to be inspected.
W-13 — Short comparison period	Fixed/adaptive comparison: 2023-01-03 to 2026-10-23.	Relatively short and potentially regime-dependent.
W-14 — Adaptive performance	No assumption that adaptive schedule outperforms.	Current results show no consistent adaptive superiority.
W-15 — Information timing	Every decision must use information available at that time.	Audit dates, target alignment, optimizer inputs, and metric windows.
W-16 — Comparable protocols	Candidate and final portfolio comparisons need compatible initialization/evaluation conventions.	Verify before combining results into a ranking.

7. Baseline Results and Interpretation Guardrails

These figures are recorded to show the context in which the assumptions operate. They are historical outputs, not forecasts.

Standard OOS (2011–2026)	CAGR	Volatility	Sharpe	Max DD	L1 turnover
Equal Weight	8.01%	11.34%	0.7359	-23.18%	6.1002
Minimum Variance	3.50%	5.26%	0.6805	-18.26%	12.7998
Maximum Sharpe	7.51%	11.16%	0.7048	-28.07%	88.8475

Stateful walk-forward (2013–2026)	CAGR	Volatility	Sharpe	Max DD	Rebalances
Equal Weight	8.21%	11.02%	0.7710	-22.53%	56
Maximum Sharpe	6.99%	11.61%	0.6403	-23.50%	127
Minimum Variance	2.78%	5.42%	0.5333	-18.96%	212

- Different backtest variants use different periods, initialization rules, and accounting details; their metrics are not automatically comparable.

- Full-sample optimized allocation metrics are in-sample.
- Results depend on the ETF universe, provider data, costs, lookback, constraints, and implementation timing.
- High historical Sharpe is not proof of a persistent edge; no strategy is established as suitable for live trading from these tests alone.

8. Transaction Costs and Market Realism

ID / topic	Assumption or decision	Status / follow-up
C-01 — Fixed bps	Costs represented as fixed basis points per unit of measured turnover.	Ignores variable spreads, volatility, liquidity, and order size.
C-02 — Zero cost	0-bps scenario is a comparison baseline only.	Not a realistic deployable result.
C-03 — Market impact	No explicit market impact model in the baseline.	Could add asset-specific spreads and turnover/ADV constraints.
C-04 — Fees/taxes	Taxes, broker fees, and fund expenses are not separately modelled in the documented baseline.	Verify how ETF expenses are reflected in price data to avoid double-counting.
C-05 — Cash yield	No explicit cash yield is added for uninvested cash.	Verify engine behavior and account for residual cash if any.

9. Code, Testing, Reproducibility, and Outputs

ID / topic	Assumption or decision	Status / implication
T-01 — Shared loader	Shared loader used to reduce inconsistent data loading.	Continue consolidating duplicate logic.
T-02 — Test scope	Tests cover engine, strategies, metrics, visualization, and stateful walk-forward behavior.	Passing tests does not prove economic validity or absence of all leakage.
T-03 — Stateful tests	Six dedicated tests cover prior-year selection, missing candidates, yearly decisions, weights/returns, interval changes, and cost accounting.	Keep them after future changes.
T-04 — Quality checks	User confirmed Ruff, targeted tests, and full pytest passed after latest stateful changes.	Re-run after future edits; applies only to tested code state.
T-05 — Outputs	Runs saved under data/runs/ and subfolders.	Use distinct run folders/metadata to avoid accidental overwrites.
T-06 — Audit artifacts	CSV outputs and charts are the main artifacts.	Add config snapshots, package versions, extraction timestamps, and code/data identifiers.
T-07 — Chart review	Chart generation succeeded; visual inspection of every chart is not confirmed.	Inspect axes, clutter, and interpretation.
T-08 — Refresh overwrite	Data refresh can overwrite existing CSVs.	Back up or version raw data before important experiments.
T-09 — Reproducibility	Depends on provider history, package versions, configuration, and any random seeds.	Record these alongside experiments.
T-10 — Version control	Git is not initialized.	Tracking exact code states is harder; reconsider when ready.

10. Machine Learning — Decisions Still Open

The ML phase has not started. The following should be decided deliberately rather than silently assumed.

ID / question	Decision to make	Research discipline
ML-01 — Target	Future realized volatility, return, drawdown risk, correlation, or another quantity?	Choose a target tied to a clear research question.
ML-02 — Horizon	5, 21, or 63 trading days?	21 days is a candidate, not a final decision.

ID / question	Decision to make	Research discipline
ML-03 — Features	Lagged returns, rolling volatility/drawdown, volume, or other data?	Use only data available at the forecast timestamp.
ML-04 — Baselines	Rolling volatility, EWMA, GARCH?	Reuse existing implementations where appropriate; inspect first to avoid duplication.
ML-05 — Models	Regularized regression or tree-based model?	Start simple; add complexity only if justified.
ML-06 — Validation	Chronological train/validation/test or rolling/expanding splits?	Do not randomly shuffle overlapping time-series targets.
ML-07 — Leakage	How to fit scaling, feature selection, and tuning?	Fit only on training history; align targets; consider purge/embargo for overlapping horizons.
ML-08 — Metrics	MAE/RMSE/QLIKE or other target-specific metrics?	Report uncertainty and regime-specific performance.
ML-09 — Portfolio value	Does better forecast accuracy improve after-cost portfolio outcomes?	Evaluate forecast quality first, then portfolio usefulness.
ML-10 — Overfit control	How to avoid tuning repeatedly to one period?	Reserve a final holdout and report unsuccessful experiments too.
ML-11 — Data scope	Retain daily ETF data or add sources?	Require credible historical coverage and timestamps for new data.
ML-12 — Integration	Should ML forecast risk or directly control weights?	Initially keep forecasting separate from allocation to diagnose any improvement.

11. Prioritized Verification Checklist

- **1. Return field and timing:** confirm adjusted-price field, simple versus log returns, timezone handling, and which day's return receives weights estimated from prior information.
- **2. Turnover and costs:** document one-way/two-way L1 turnover, cost mapping, and consistency across scripts.
- **3. Optimizer details:** record constraints, objective, risk-free rate, covariance estimator, solver, tolerances, and fallback behavior.
- **4. Walk-forward audit:** verify all decisions use only prior information and compare equivalent initialization/evaluation rules.
- **5. Universe and data limitations:** check adjusted prices, inception dates, missing dates, and survivorship/availability concerns.
- **6. Reproducibility:** save configuration, provider timestamp, package versions, and a data snapshot/checksum.
- **7. ML research question:** choose what the forecast should answer before choosing a model.
- **8. Naming/version control:** finalize the name only when useful; consider Git deliberately without prematurely changing project structure.

12. Compact Summary

The current system assumes daily ETF returns, a fixed ten-ETF universe, a 252-trading-day estimation window, a 21-day standard rebalance schedule, a 10-bps proportional cost model, long-only fully invested allocations, 252-day annualization, and a zero risk-free rate for the documented Sharpe calculations. Robustness experiments vary rebalance intervals and costs. Walk-forward methods select schedules from historical performance, while the stateful simulator carries holdings and drift through time.

> These are research conventions, not universal truths. The highest-priority unresolved issues are return timing, turnover/cost semantics, optimizer specification, and a source-level audit of all walk-forward scripts for leakage. ML target, horizon, features, model, validation, and portfolio integration remain undecided.

End of register — 28 October 2026